Predicting mold risk in the building stock of NYC and its impact on the incidence of respiratory illnesses

Final Project Report ML for Cities
May 5th, 2023

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Github Repository:

https://github.com/bbonczak/Predicting-mold-risk-in-the-building-stock-of-NYC.git

Datasets on Google Drive:

https://drive.google.com/drive/folders/15RhTrF5C2_VZF66hZLeU9fZ6N5u9DYHu?usp=sharing

Abstract

Exposure to mold can trigger or worsen asthma symptoms and allergies. Mold detection in New York City is done through self-reporting 311 complaints and mold violations, as reported by the city's Housing Preservation and Development from biyearly inspections. The union of these datasets, although aid in characterizing at-risk mold buildings, underreport mold cases in NYC. Therefore, they do not establish the ground truth. To understand the mold-risk index of every building of NYC, this study employed a Random Forest model to classify buildings based on their risk of mold infestation. NYC Primary Land Use Tax Lot Output (PLUTO) data was used to physically characterize buildings and 500-year flood risk data from NYC Planning helped in accounting for humidity conditions. A mold risk index for each residential building was calculated based on the at-risk classification data and used as a predictor variable along with US Census data on demographics and socioeconomics as control variables. To explore the relationship between asthma and mold an ElasticNet regression model was employed. With mold risk index, the regression model accuracy improves significantly, indicating the veracity of the classification model.

1. Introduction

Dampness and moisture in buildings increase mold growth risk. Mold spores negatively affect indoor air quality, which, in turn, has specific health outcomes for sensitive individuals. It has been repeatedly proven that exposure to mold spores is heavily linked to the prevalence of asthma symptoms in humans and, in some extreme cases, is directly associated with the patient's death (Moses et al., 2019; O'Driscoll et al., 2005; Targonski et al., 1995).

To better monitor mold issues and reduce its health impact, in New York City (NYC), the City Council established Local Law 55 in 2018 to require landlords "of buildings with three or more apartments or buildings of any size where a tenant has asthma to keep tenants' homes free of mold," and to repair the conditions that cause these problems (Indoor NYC Housing Preservation and Development (HPD), 2018*)*. If a building does not comply with LL55, HPD issues a violation for this building, and this record is publicly available via NYC Open Data. In addition to city-issued violations, residents can use the NYC 311 system to self-report any mold incident and wait for HPD to inspect the issue. The city also closely monitors asthma-related emergency room visits and hospitalization. The NYC Department of Health (DOH) reports asthma as "a leading cause of emergency room visits, hospitalizations and missed school days in New York City's poorest neighborhoods" and recommends mold prevention as a crucial step to reduce asthma health risks (NYC DOH, 2023).

While NYC takes the initiative in addressing mold issues and reducing its health impact, this current process of identifying buildings at risk of developing mold is predominantly a manual process with visual inspection as the primary method, which can be time-consuming, subjective, and labor-intensive Center for Disease Control and Prevention (CDC) (2022). Therefore, it is crucial to develop efficient methods for early detection in buildings in the interest of public health to reduce the health risks associated with mold growth.

With the limitations of manual methods for identifying buildings at risk of developing mold, there has been a growing interest in using machine learning (ML) models for automated mold risk assessments. Recent studies have shown that ML algorithms can effectively predict indoor mold growth and identify at-risk buildings. For example, Hegarty et al. (2020) applied a random forest (RF) model to classify the mold status of a home based on the DNA sequences found in settled dust samples. The samples were collected from homes in various climates across the United States. The RF model used default parameters implemented in R. Another recent example of ML application for identifying mold contamination is the study by Łagód et al. (2017), in which the authors used electronic sensors to collect indoor air samples and performed classification for the sensor readouts with principal component analysis. The same group published another study (Majerek et al., 2020) that changed the classification method to a support vector machine, followed by one more (Majerek et al., 2021) that used DBSCAN for classification.

While the studies above showed potential improvement in identifying mold contamination compared to pure visual inspection, these models require professional training for data collection, and the ML methods were not necessarily tuned to the best performances. In addition, these models require a sample collected directly from the location to identify whether a given building has mold. In contrast, we aim to develop an ML approach that uses existing, publicly available data to detect mold growth in buildings and tune the model for best performances. We divide the study into three parts, starting with an RF classification to identify a building as having mold or no mold. Compared to the literature cited above, aside from avoiding the extensive data collection procedure, another potential advantage of our classification model is using feature similarity with labeled training data to detect building mold risk early. Then, we develop a mold risk index based on the results from the classification model. The last step is performing a regression analysis of our index with demographic and health indicators.

2. Data and Methods

2.1. Datasets and data processing

For the purpose of this study we utilized several data sets to inform the analysis and validate our findings. The summary of the data used is presented in Table 1 followed by the more in depth description and processing steps.

Table 1. Data summary

Name	Source	Temporal Coverage	Geographical Unit	Size
Mold Violations	NYC HPD Housing Maintenance Code (HMC)	2010 - 2022	Residential Unit	214,881
311 Mold Complaints	NYC 311 Service Requests	2010 - 2022	lat, lon	312,085
Primary Land Use Tax Lot Output (PLUTO)	NYC Department of City Planning	2020	Borough-Block-Lot (BBL)	858,619
500-yr Floodplains	NYC Mayor's Office of Climate and Sustainability	-	Citywide	-
American Community Survey	US Census Bureau	2019	Census Tract	2,165
Annual Average Asthma Rate	NYC Department of Health	2017-2019	Neighborhood Tabulation Area (NTA)	188

2.1.1. Mold Violations

The NYC HPD issues rental dwellings and buildings violations regarding the New York City Housing Maintenance Code (HMC) or the New York State Multiple Dwelling Law (MDL). We used the mold violations from 2010 to 2022 obtained from NYC Open Data (NYC HPD, 2023). The data provides information on the date and type of violation issued, specific location and description of the problem, along with its severity and resolution. During the study period, there were 214,881 mold violations issued for 33,905 unique properties identified by the Borough-Block-Lot (BBL) number.

2.1.2. 311 Mold Complaints

We used the 311 Service Requests related to mold as a complementary indicator for the potential mold occurrence in the building (NYC Department of Technology and Information Telecommunication, 2023). We obtained this data through the NYC Open Data Platform. For the purpose of this study we match the study period to the mold violation data, namely 2010-2022. Similarly to the mold violation data, this data set contains information about the reported complaint's date, type and geographic coordinates. The original data contained 312,085 complaints. In order to make this data compatible with the violation data set, we performed a spatial join using Geopandas Python package to the Primary Land Use Tax Lot Output (PLUTO) data (described below) to obtain BBL identifier of the complaint. We then grouped all of the complaints occurring during the study period on the BBL level, resulting in 56,072 unique BBLs affected.

2.1.3. PLUTO

We used the Primary Land Use Tax Lot Output (PLUTO) as the basis for the building-level mold analysis identified by BBL number. (Department of City Planning (DCP), 2020) This data set was obtained directly through NYC Open Data API. It contains geometries of tax lots in New York City, as well as plethora of other information about the city properties, including address, zoning, land use, building type and physical characteristics such as size, age, number of units and assessed value etc. The original data contains 858,619 records and 90 columns. BBL identifier was used to map both mold violation and complaints occurrences For the purpose of mold classification we decided to focus on building specific characteristics reported in PLUTO such as location, year of construction and / or alteration, building class, height, gross floor area (GFA), number of units, number of floors, information about the building shape, basement, location within the neighboring properties and assessed value. We then performed additional feature engineering to calculate property age, time from most recent renovation, proportion of GFA used for residential purposes, average unit size, building shape ratio and average value per square feet. We also generated binary variables for main building types, presence of basement and whether it is a standalone building. After processing, the resulting data consists of only 36 selected and generated columns.

2.1.4. Floodplains

We decided to incorporate floodplain intersection as an additional attribute in mold detection because buildings in a floodplain likely have humid living conditions, increasing the risk of mold. The linkage between flooded areas and mold-induced health issues is widely reported. For example, residents in communities most affected by Superstorm Sandy developed asthma or other respiratory symptoms caused by improper post-storm cleanup (Jeffrey-Wilensky, 2022). The 500-yr floodplain designates VE, AE, AO zones with severity ranges from high to moderate. We obtained the shapefile from NYC Open Data, removed the rows with no flood zone designation and spatial-joined the cleaned data to PLUTO and added a binary column to indicate whether the PLUTO point is in the floodplain.

2.1.5. ACS Census Data

We used 2019 US Census Bureau data to control for demographic characteristics that could have casual relationships with asthma risk. We used age and sex, race, household income characteristics, poverty index data to characterize age and sex distribution, economic conditions, and housing conditions of New Yorkers. Demographic data were collected from the US Census Bureau at the census tract geographic unit level and aggregated to the Neighborhood Tabulation Area level (NTA) from 2010.

All demographic and socioeconomic data used in this study were extracted in the census tract geographic scale. While socioeconomic indices were downloaded using an API (https://api.census.gov/data/2019/acs/acs5/subject/variables.html), age and sex data were procured through manual web download due to the lack of access to this data through API. With the goal to aggregate the datasets to a Neighborhood Tabulation Area (NTA), a spatial join operation was performed after ensuring that the shapefiles from the two geographical scales were on the same Coordinate Reference System (CRS): GPS-EPSG:4326 based on the WGS84 datum.

For the spatial join operation, the census tract level data were converted into centroids(points) and joined to the NTA shapefile (polygonal). Ultimately, the data was aggregated to the NTA level using suitable summary statistics (mean, sum as applicable). It is worth noting, the challenge in this step lay in dealing with overlaps and discrepancies between the boundaries of census tracts and the NTAs. It was an important consideration for avoiding double-counting or other errors that would lead to bias in the analysis.

2.1.6. Asthma Cases

The New York City Department of Health provides asthma-related emergency department (ER) visits, hospitalization, and prevalence on the Environmental and Health Data Portal (NYC DOH, 2023). This study focuses on the Neighborhood Tabulation Area (NTA) level ER visits, which offer the highest spatial resolution and most recent data from the DOH portal. While the portal provides data for three age groups (i.e., under 4 years old, 5-17 years old, and above 18 years old), the temporal availability does not match up among the three groups, as shown in Table 2.

Table 2. Data availability for asthma-related ER visits at NTA level

Age group	Years available
Under 4 years old	• 2014-2016
5-17 years old	• 2014-2016 • 2017-2019
18 and above	• 2012-2014 • 2017-2019

To determine which files to use for our analysis, we performed two-sample ks-tests on the following pairs: a) 2017-2019 age-adjusted average annual rate per 10,000 for adults and average annual rate for adolescents; b) 2014-2016 average annual rate per 10,000 for adolescents and young children. We chose the age-adjusted rate for adults provided by the city since the adult asthma data covers everyone above 18, and asthma affects young adults differently than the older population. The city did not report age-adjusted rates for the two age groups of children.

The p-value from the first test was nearly 0, suggesting that we could reject the null hypothesis that the distribution of the 2017-2019 annual average rate of ER visits is the same for adults and adolescents. While the p-value from the second test was 0.04 (i.e., lower than the significance level at 0.05), to include as many age groups as possible, we treated this result as suggesting the distributions of the 2014-2016 annual average rate of ER visits for young children and adolescents were the same, and we could use the data for adolescents to represent all New Yorkers under 18. However, it is worth noting that this approach assumes that asthma outcomes amongst young children stayed the same from 2014-2019 and that the spatial distributions of asthma rates of young children and adolescents are comparable. In summary, we selected the 2017-2019 datasets for adults and adolescents (Figure 1) as our metric for representing asthma health outcomes.

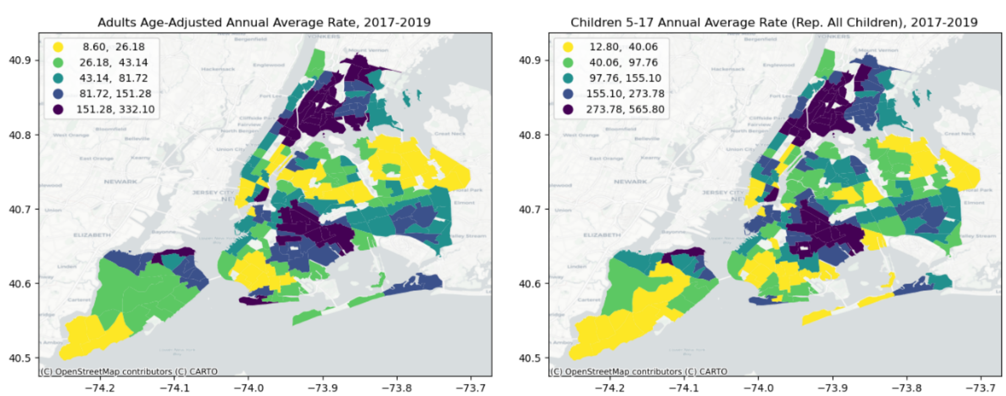

Figure 1: Asthma Rates for Adults and Children

2.2. Methods

In this section we describe the methods used in the study, as visualized in Figure 2. After initial data collection and processing, we want to use ML classification techniques to identify buildings at-risk of developing mold based on their structural characteristics. Secondly, we develop a Mold Risk Index (MRI) at the Neighborhood Tabulation Area (NTA) level. Finally, considering the research on causal relationship between mold and respiratory disease risk, we use the calculated MRI to investigate the correlation between the predicted at-risk mold areas and asthma cases as a way to validate our classification findings.

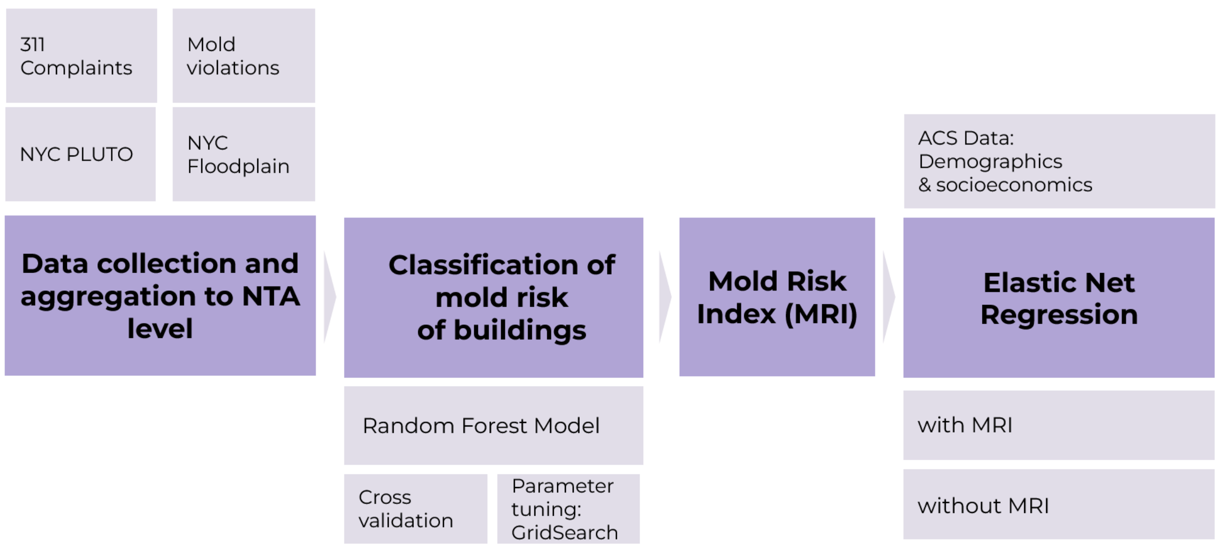

Figure 2. Study Workflow Overview

2.2.1. Classification

Our core analysis is to classify NYC buildings based on the risk of mold occurrence. To achieve that, we use Random Forest Classifier implemented in the scikit-learn package for Python. (*scikit-learn*, 2023) Random Forest (RF) is an ensemble machine learning method that learns from several decision trees, where each tree in the ensemble is built from a sample drawn from the training set. The purpose of introducing randomness is to decrease the variance of the forest estimator.

To identify buildings at risk of developing mold, we first prepare the training data. To do this we combine processed mold violation and 311 mold complaints data with building level characteristics derived from PLUTO and the binary information whether the property is located within the floodplain. All four (4) data sources are merged on the BBL unique identifier. The resulting data consists of 858,619 properties, out of which only 33,849 (4%) had mold violations and 55,760 (6.5%) had mold-related complaints between 2010 and 2022. Because of the high overlap between the properties with complaint and violation (28,753 or 85% of all violations) we decided that we consider the existence of any of those cases as a composite “mold issue”, denoted as “1”, which occurs in 7.1% of all properties. The remaining observations were considered mold-free and labeled as “0”. This becomes our dependant variable for the remainder of the study.

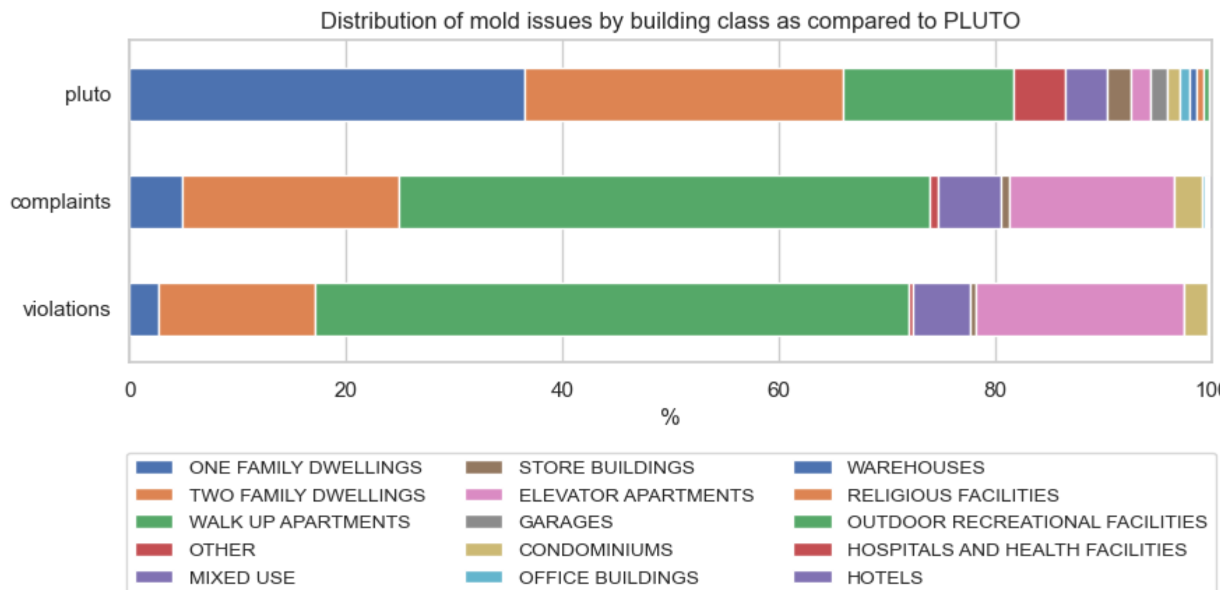

Figure 3: Distribution of mold issues by property type

As illustrated in Figure 3, the properties reported mold complaint or received mold violation do not represent the overall building stock in New York City. Those types of issues tend to be more frequently reported in larger, multi-family residential properties, especially in walk-up and elevator apartments and are significantly underrepresented among the single family dwellings. It is not surprising, given the ownership vs. tenant structure of those types of properties. Nevertheless, since almost all of the mold-related issues occurred in residential buildings, we decided to focus our analysis only on those types of properties. The final dataset used for modeling, after selecting residential properties, removing missing values and dropping duplicates consists of 749,383 observations and 29 features. The list of features used and pairwise correlation between them is illustrated by the correlation matrix presented in Figure 4.

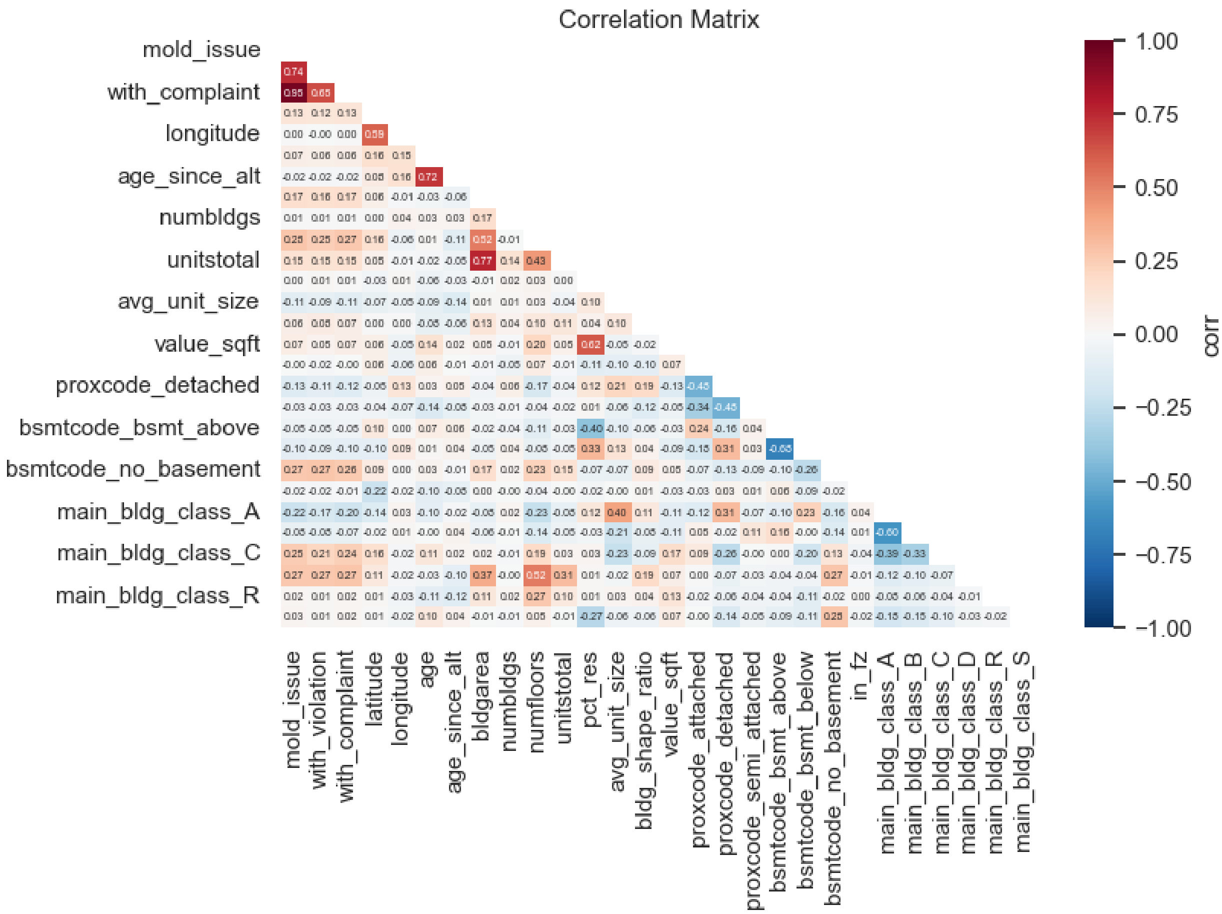

Figure 4: Feature correlation matrix

We train the Random Forest classifier predicting building-level mold risk in three stages:

1. <u>Exhaustive Grid Search Cross-Validation</u>

We performed an exhaustive hyper-parameter tuning using grid search and 5-fold cross-validation using 0.3 train-test data split and assessed the resulting ROC AUC score. After approximately 3,000 interactions we achieved the best estimator with the following set of parameters: max_depth=100, max_features=0.25, min_sample_leaf=50, n_estimators=200, criterion='entropy', max_samples=0.5. The Out-of-Sample (OS) ROC AUC was 0.881.

2. <u>Class re-balancing</u>

Since our data is highly imbalanced (mold occurrences account for only 7% of building stock), the resulting model had a dangerously high False Negative Rate (FNR). Therefore we applied class re-balancing by using the class_weight parameter. The optimal parameter was 1:11 for mold occurrence. The resulting model achieved slightly higher OS ROC AUC of 0.882, but significantly lower miss rate, at the cost of higher False Positives Rate (FPR).

3. <u>Sensitivity Adjustment</u>

Finally, to further reduce the type II error, we performed sensitivity adjustment by optimizing the probability threshold at which to consider buildings at-risk of developing mold keeping other model parameters constant. The probability threshold with the highest ROC AUC value was found at 0.47, which is illustrated in Figure 5.

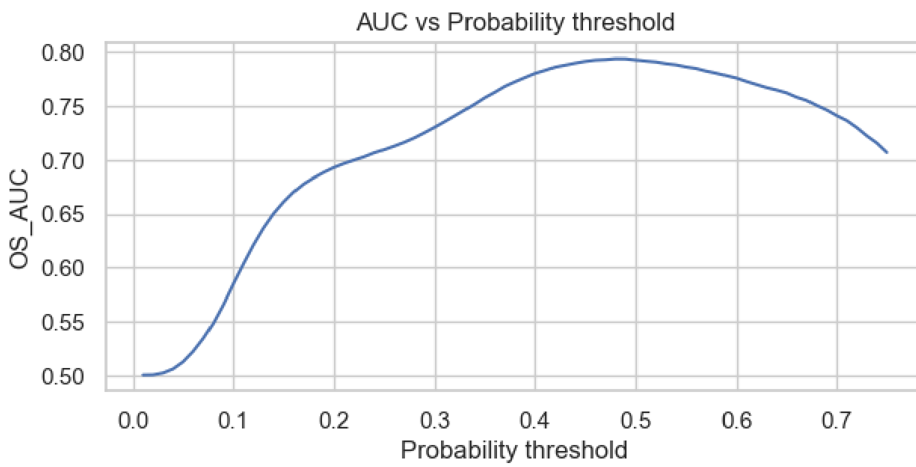

Figure 5: RF model sensitivity adjustment

Final RF classification model was then applied to the entire data set of all 750 thousand residential properties in New York City.

2.2.2. Mold Risk Index

The classification result provides an indication of the probability of mold occurring on a building level. For the purpose of tying the classification in with other data for further analysis, it is necessary to establish a method for aggregating the classification results on NTA level. Simply counting the number of positively classified buildings per NTA would ignore the variation in building sizes within each zone and its potential impact on residential population.

Moreover, mold has a significant impact on people who frequently return to mold-infested indoor areas. Therefore, we opted to use the residential floor area as a normalization measure to calculate the MRI. Consequently, the Mold Risk Index is computed as the proportion of positively classified residential floor area in square feet divided by the total residential floor area summed for each NTA.

2.2.3. Regression

As discussed earlier, exposure to mold increases asthma health risks (Moses et al., 2019; O'Driscoll et al., 2005; Targonski et al., 1995). Therefore, we assume the previously computed Mold Risk Index (MRI) should influence the confirmed asthma cases per NTA.

To quantify the relationship between the MRI and the asthma rates, as well as assess the influence of the MRI against demographic, race, and income-related information, a linear regression method is implemented. The regression additionally fits the purpose of assessing whether the classification can be seen as being trustworthy. Depending on the influence of the MRI on Asthma Rates, a causal relationship to the accuracy of the classification can be made.
The input data required for the regression is compiled through ACS Census Data for 2019, the MRI and the asthma rates for children and adults on the NTA level, which have been processed as described in section 2.1. While the complete number of NTA's in NYC is 195, the input dataset to the regression here contains information for 141 NTA's over 38 columns. This is potentially due to the geometry of census tract centroids and that of NTA not matching during the spatial join.

Given the relatively high dimensionality compared to the sample size, it has been decided to apply a regularized linear regression method to assess the influence of the MRI. The choice fell on ElasticNet-Regression (scikit-learn, 2023), as it combines the penalization parameters from Ridge and Lasso Regressions, leading to increased accuracy-robustness through weighing and selecting features according to their importance on the prediction of the dependent variable, as well as being proven to handle correlated data better than other Regression methods (Schreiber-Gregory & Jackson, 2018).

Nevertheless, to optimize the ElasticNet regression result, various feature engineering techniques were applied that reduced redundantly represented information in various features, thereby reducing overall collinearity while maintaining interpretability. Additionally, the independent variables have been standardized.

In Figure 6 the resulting correlation matrix can be seen. The applied feature engineering techniques proved successful in minimizing the correlation between the attributes. The overall mean correlation coefficient for the processed features is 0.066, representing the input data's reasonably low correlation level. However, strong correlations were still found, especially for the column indicating the proportion of people living in poverty ('poverty_status_percentage') when compared to the Total population ('d_Total population') with a coefficient at -0.69, and the proportion of working population ('total_workers_percentage') with a coefficient at 0.93.
Despite these remaining correlations, ElasticNet is expected to handle them effectively through the automated regularization of the input variables.

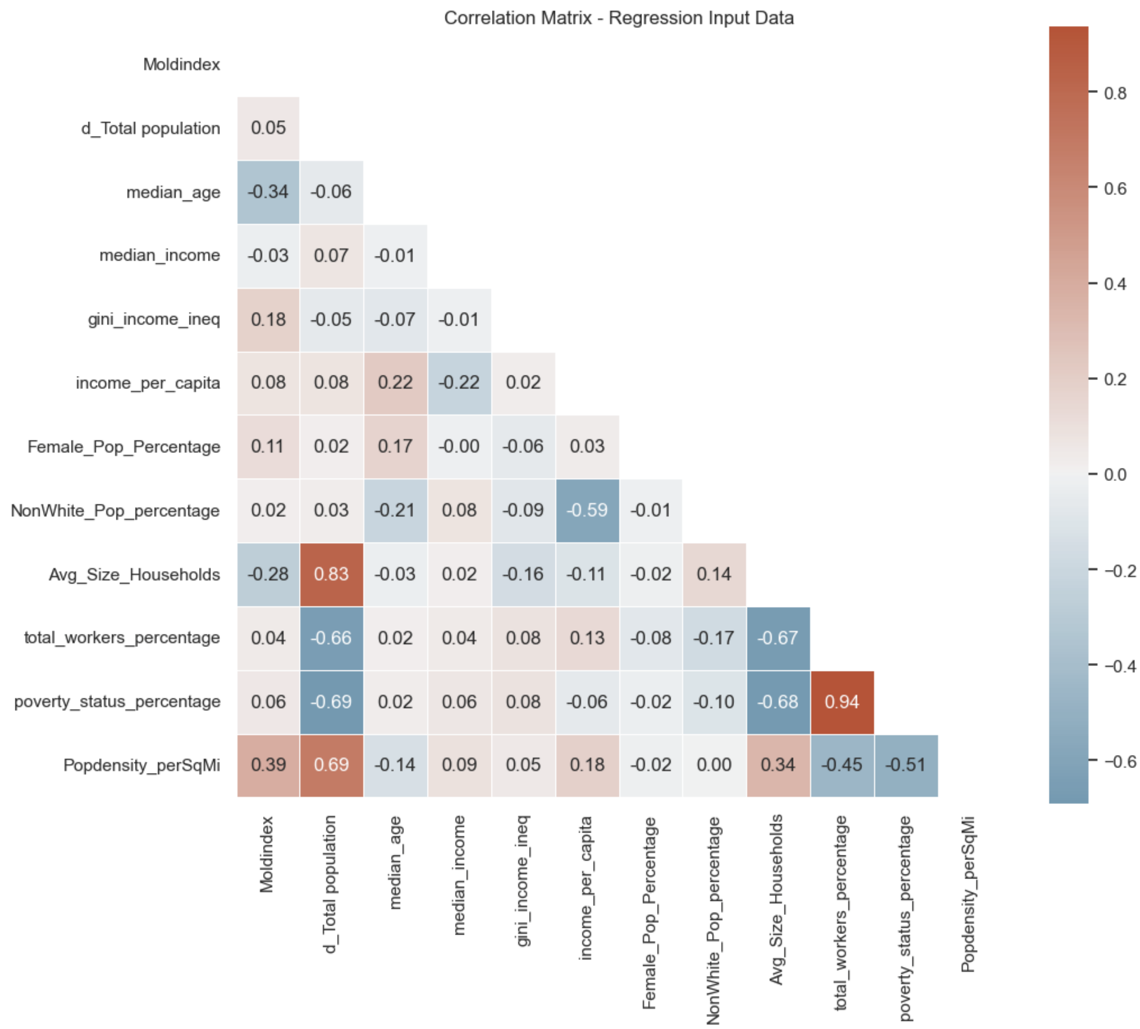

Figure 6: Correlation matrix for the Regression input data

Upon modifying the features, the data was split into training and testing samples with a ratio of 4:1. This decision was made intentionally to allow the model to achieve a comprehensive fit given the limited number of NTA's while preserving a sufficiently large portion of the data to assess the model fit. To allow for a consistent assessment of the regressions, the randomization factor of the splitting procedure has been set to a uniform factor, ensuring that all models utilize the same data for training and testing.

With the regression models, we aim to analyze the influence of mold prevalence on children's asthma rates and age-adjusted adult asthma rates. Since these two rates differ significantly, with children having higher asthma rates per NTA on average, a separate regression is implemented for each group. Further, it is intended to analyze whether the accuracy of the regression increases or decreases when the MRI is excluded from the set of independent variables for each group. In total, we will set up four regression models: two for asthma rates among children (including and excluding the MRI) and two for asthma rates among adults (also with and without the MRI).

An important Hyperparameter significantly controlling the outcome of ElasticNet Regressions is "alpha", which controls the balance between the weighting of L1 and L2 regularization, which correspond to the penalties applied in ridge and lasso regressions. This parameter is chosen based on a Gridsearch, with a scoring method maximizing R^2. Additionally, it's been chosen to cross-validate the regression 5 times, to achieve a more accurate estimate of each model's performance.

Once each regression model has been computed, the out-of-sample R^2 value is reported, along with the feature importance and the optimal alpha parameter selection. Feature importance is visualized through a horizontal bar plot that displays the regression coefficients, indicating the positive or negative influence of each variable on the prediction of asthma rates. The optimal alpha parameter selection is shown using a line plot, indicating whether the model was able to identify a specific value that yielded the best results.

3. Results

The final classification Random Forest model achieved an ROC score of 0.873 predicted on the entire data set. More detailed summary of the model performance is shown in Table 5. The overall accuracy of the model was 86.4%, with TPR of 71.9%. With the parameter tuning, re-balancing and sensitivity adjustment, we were able to reduce the FNR to 18.1%. We would like to point out the seemingly high number of instances where our model identified buildings at risk of mold infestation without confirmation in the underlying data. With 85,453 occurrences, leading to a high FPR value of 11.4% it is more than 32% higher of all mold occurrences reported. However, those cases might in reality be a valuable insight for the city administration pointing out to which properties might have mold problems, but are not currently monitored by any city agency nor their residents are willing to engage in the reporting system such as 311.

Table 5. Confusion Matrix for RF Classification.

N = 749,383 TPR = 71.9% FNR = 18.1% FPR = 11.4%		**Predicted**	
		Mold	**No mold**
Actual	**Mold**	41,809	16,377
	No mold	85,453	605,744

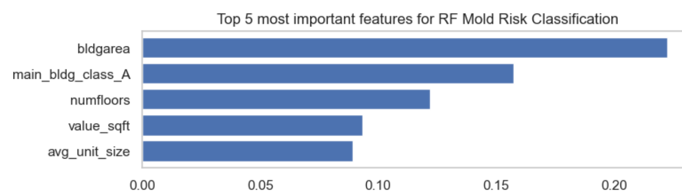

Figure 7: Top 5 most important RF classification features

Further investigation of the model shows that the most important features that model selected for deciding whether the property is at risk of mold are related to building size (as represented by the Gross Floor Area ("*bldgarea*") and number of floors). Unsurprisingly, the value of the property, which might be an indicator of the quality of building management is also significant. Top 5 most significant features picked up by the model are illustrated in Figure 7.

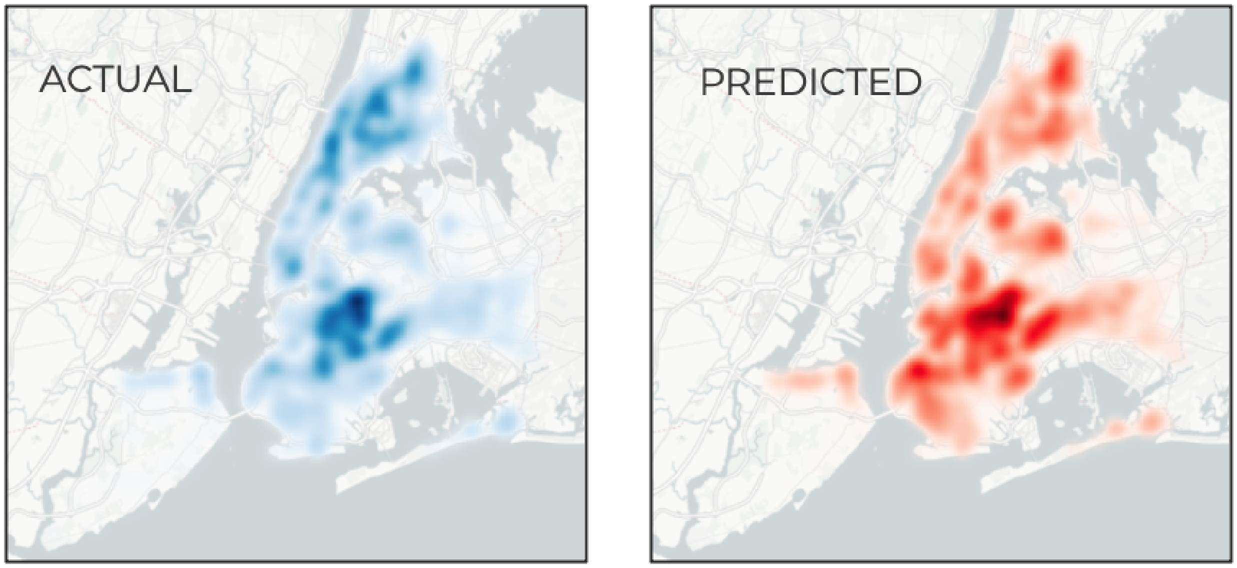

Figure 8: Spatial distribution of buildings with verified mold problems (left) and identified by classification model as at risk of mold occurrence (right)

Finally, when evaluating the spatial distribution of the actual and predicted occurrences of mold in New York City, as depict in Figure 8, one can observe that the overall pattern is similar with the same hotspots around Bushwick and Ridgewood at the border of Brooklyn and Queens, Crown Heights neighborhood of Brooklyn or Wakefield in The Bronx and Lower East Side of Manhattan. Interestingly, the classification model highlights potential mold-related problems in the areas of Astoria and Jackson Heights (Queens), Gowanus, and Borough Park in Brooklyn, which are not reported by the official statistics.

With currently used data, it is impossible to verify whether the overestimation of the predicted building-level mold occurrences are related to actual mold problems. In absence of the building inspection data and complaints records we decided to use the asthma ER visits rates as a proxy. In order to bring the building-level mold prevalence into the NTA-levels of asthma data, we developed the NTA-level Mold Risk Index described in the Methodology section. Average MRI for New York City is 0.62 with the highest being in Stuyvesant Town in Manhattan and West Brighton in Southern Brooklyn. Areas of the lowest MRI are the NTAs of lower densities of housing in Staten Island and Eastern Queens. Overall, the borough with the highest average MRI rates is Manhattan (0.88), but it is driven mostly by high density residential areas of Harlem, Washington Heights and Inwood, similar to neighboring parts of the Bronx. Figure 9 visualize the distribution of the MRI within the city.

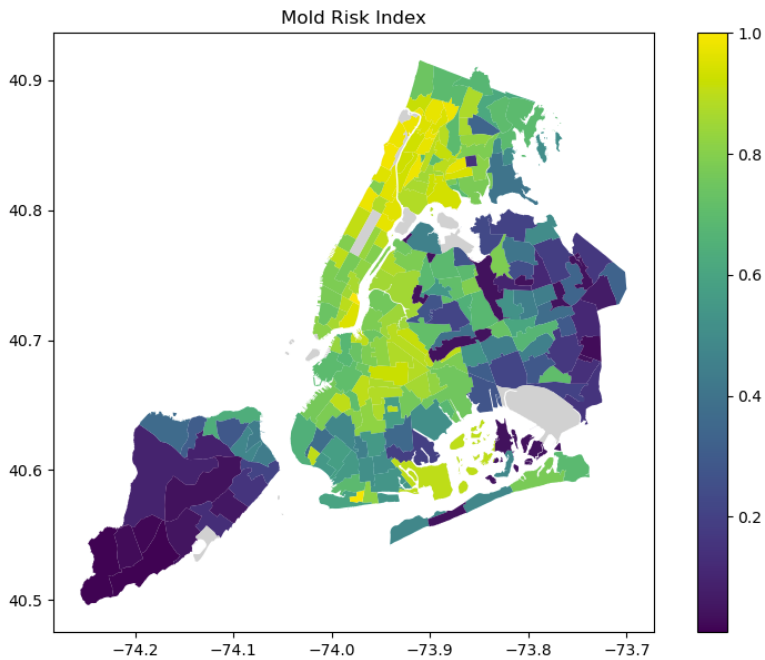

Figure 9: Mold Risk Index

In Table 3 the accuracies of the regressions with the different input data constellations can be seen. It is evident that the accuracies for the regressions that include the MRI are higher.
The feature importances for the regressions that include the MRI as an independent variable are shown in Figure *10.* The regressions without the MRI are shown in Figure 11.

From these figures it can be seen that for both adults and children the Mold Risk Index has a high positive influence on the incidence of asthma rates. For the Regressions without the MRI the importance has a different ranking, however an observation for both cases is, that the asthma rates are majorly driven by the proportion of diverse populations, indicating the fact that minorities are more likely to be suffering from asthma which is confirmed by literature (Forno & Celedon, 2009).

Due to the causal link between asthma incidences and mold spore exposure, the increased accuracy and high importance of MRI within the regressions confirms the truthfulness of our classification result.

This is especially interesting given the amount of buildings that have been classified positively for mold, but have no history of mold exposure through 311 service calls or recorded violations. Therefore this group of so-called false positives, represents the buildings where mold is likely to occur in future, and it is advisable to conduct inspections for mold from the authorities in charge.

Regression Input	**Accuracy Score R^2**
Children Asthma Rate with Mold Risk Index	0.642
Children Asthma Rate without Mold Risk Index	0.538
Adults Asthma Rate with Mold Risk Index	0.675
Adults Asthma Rate without Mold Risk Index	0.594

Table 6: Regression Accuracy Scores R^2 for different Asthma and Mold Risk Index Input Data

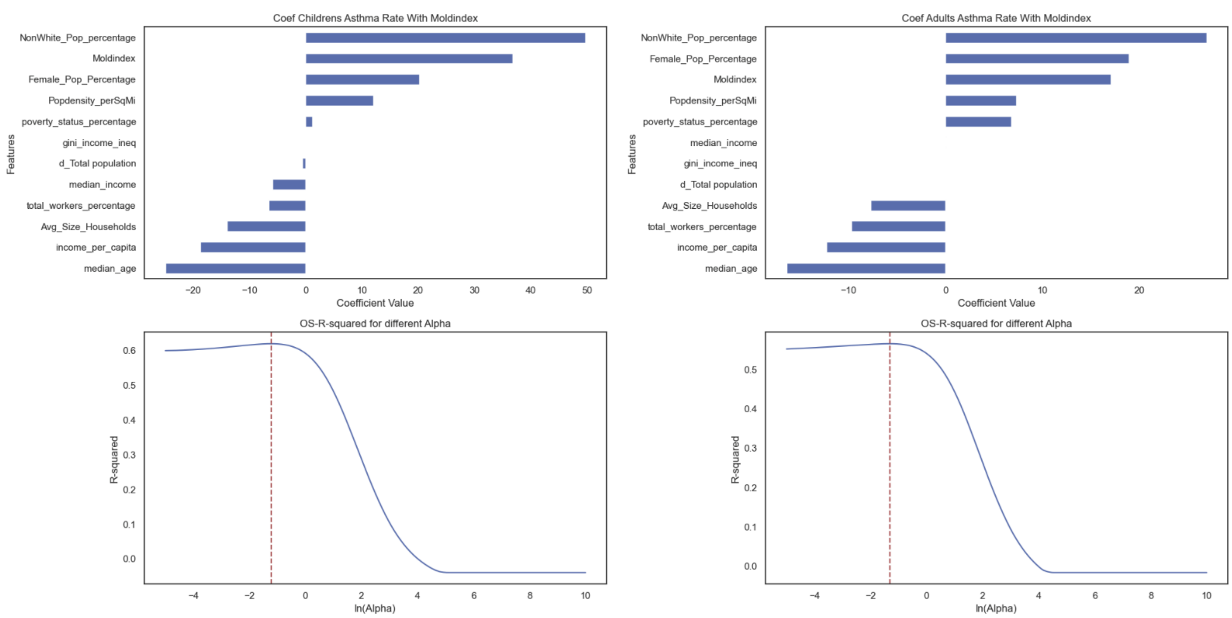

Figure 10: Regression Feature Importance with Mold Risk Index

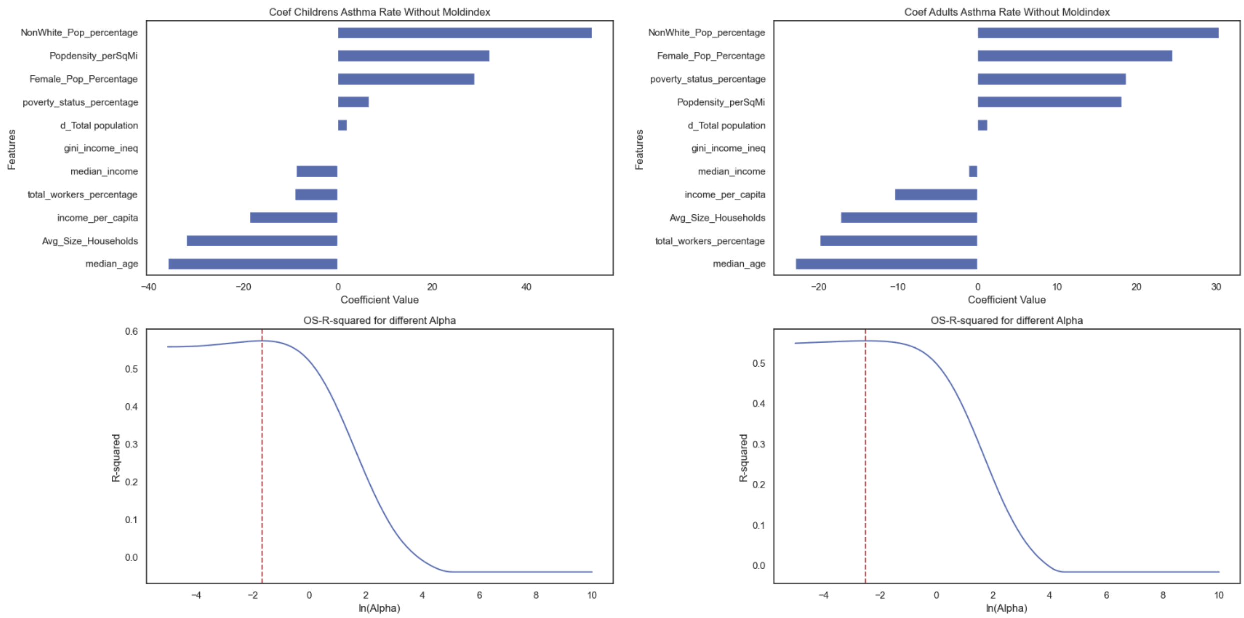

Figure 11: Regression Feature Importance without Mold Risk Index

4. Discussion

4.1 Discussion on Results

Overall, the project addressed an important urban issue using different ML techniques such as classification and regression. Firstly, the project showcased a classification-based method to identify at-risk target areas where establishment of ground truth is not feasible, making this a useful tool for policy-makers to address urban issues proactively rather than reactively. Secondly, the project built a trustworthy workflow that can statistically illustrate the relationship between asthma rates and mold risk index. The regression model performed better with the Mold Risk Index as a predictor variable, demonstrating the effectiveness of the study design. Lastly, and most importantly, the efficacy of the regression model shows the potential need for reevaluation of data collection techniques by the city for mold identification in buildings and shows the potential of using ML techniques in identifying and mitigating urban issues.

4.2 Limitations and future directions

The first limitation of this study lies in the data aggregation method (census tract- centroid- NTA) used for ACS data. This leads to discrepancies in data allocation: some NTAs are overallocated with data while others underreport data or have no data allocation. To circumvent this problem, we recommend using more sophisticated spatial techniques such as partial weighting or interpolation, which may be more suitable.

Additionally, to further this study, we recommend using more datasets in both the classification and the regression models. For the classification model, using 311 Water Leak data may be a more appropriate indication of localized dampness issues in buildings- a direct cause for mold infestation in buildings, as well as indication of overall building quality condition (NYC Health, 2023). For the regression model, we suggest using additional socioeconomic attributes such as Health Insurance coverage and Occupation.

4.3 Policy Recommendations

The ML-based classification tool using a class rebalancing technique may be a useful tool for NYC's Department of Housing Preservation and Development to streamline their workflow in identifying buildings at-risk mold building, aiding in proactive decision-making. Additionally, the at-risk mold model may help in the city's budget allocations.

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6. Member Contribution

Table 4: Member Contributions

Name	Contribution
Bartosz Bonczak	Mold Violation and PLUTO data collection and processing, Random Forest Classification
Sreya Chakraborty	ACS data collection and aggregation to NTA
Matthias Fitzky	311 Data Collection, Mold Index Calculation and ElasticNet Regressions
Enola Ma	Floodplain designation for PLUTO and asthma data collection
All Team Members	Project ideation and scoping, data mining, presentation and report preparation.